The empirical recurrence for OEIS A235785: recovering a conjecture that is not written down

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Abstract

OEIS A235785 records "Empirical recurrence of order 49", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 71$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 49, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 49 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235785 is "Number of (2+1)X(n+1) 0..2 arrays with the difference of the upper and lower median value of each 2X2 subblock in lexicographically nondecreasing order columnwise and nonincreasing rowwise.". It has offset 1 and begins

$$537, 7331, 86691, 922543, 9242949, 88065309, 811287445, 7265948995, \ \ldots$$

The entry states:

> Empirical recurrence of order 49 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:49:40, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 188$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^{\top} M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 71$ of the 188, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 376$ exact terms of the model returns order 49 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{49} c_i\, a(n-i),$$

c_1	24	c_{14}	−517376664	c_{27}	−288705962443592	c_{40}	478866379358208
c_2	−114	c_{15}	169131083293	c_{28}	−1309176227296032	c_{41}	163761594662912
c_3	−1548	c_{16}	−142366921164	c_{29}	1060047533812048	c_{42}	−160589186334720
c_4	15712	c_{17}	−1155185014461	c_{30}	1857566495018528	c_{43}	−4668431400960
c_5	20122	c_{18}	1803988176482	c_{31}	−2266130390102656	c_{44}	26685548789760
c_6	−703112	c_{19}	5622676581663	c_{32}	−1734046794577088	c_{45}	−3263777734656
c_7	1116123	c_{20}	−13162054270937	c_{33}	3244995367830400	c_{46}	−1900169658368
c_8	16697633	c_{21}	−18756320795936	c_{34}	789794221720064	c_{47}	375700586496
c_9	−56064091	c_{22}	66282149310044	c_{35}	−3172157079172608	c_{48}	41691381760
c_{10}	−232001700	c_{23}	36152864015582	c_{36}	292357800871936	c_{49}	−10066329600
c_{11}	1256330943	c_{24}	−242901721992520	c_{37}	2056573736775680		
c_{12}	1721960413	c_{25}	4336406393812	c_{38}	−709304009265152		
c_{13}	−17597372414	c_{26}	657900211481032	c_{39}	−817248347160576		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 49, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $49 + 4$ terms removed returns order 49 as well, so $\deg q_{\min} = 49 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $49+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 49 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A235785.

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